

MARACAY, Venezuela

WMO: 804130

Lat: 10.250N

Lon: 67.649W

Elev: 408

StdP: 96.52

Time Zone: -4.00 (W04)

Period: 83-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	18.4	12.9	9.8	28.5	14.1	10.5	27.6	7.5	26.2	6.4	27.0	0.6	50	0.242

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.5	34.8	23.9	33.9	23.6	33.1	23.4	26.8	31.0	26.1	30.6	25.6	30.1	3.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	22.3	29.0	25.1	21.2	28.2	24.4	20.3	27.6	87.0	30.9	83.7	30.9	81.3	30.2	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.2	6.0	5.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	25.3	26.6	27.2	27.4	27.1	25.9	25.3	25.4	25.8	26.2	26.0	25.4
	DBStd	1.39	1.15	1.37	1.46	1.21	1.29	1.25	1.08	1.17	0.95	0.96	1.00	1.14
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5889	473	464	532	522	531	478	475	479	474	501	481	479
	CDD18.3	2848	215	231	273	272	273	228	217	221	224	243	231	221
	CDH23.3	25163	2005	2378	2868	2790	2628	1873	1533	1630	1761	1920	1863	1913
	CDH26.7	10016	841	1107	1358	1223	1043	646	481	534	620	703	706	754
Wind	WSAvg	1.9	2.1	2.3	2.4	2.2	2.0	2.0	1.8	1.7	1.6	1.5	1.5	1.8
Precipitation	PrecAvg	940	4	4	8	46	100	130	136	166	136	105	60	15
	PrecMax	1597	35	38	70	198	337	409	454	362	271	201	221	78
	PrecMin	374	0	0	0	0	8	7	39	62	29	34	2	0
	PrecStd	244	8	8	15	47	57	76	75	68	63	37	42	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	35.8	36.0	35.1	34.7	33.9	32.1	32.6	32.7	33.2	33.2	33.1
		MCWB	22.3	23.4	23.0	25.2	24.0	24.3	22.9	24.0	24.6	25.0	24.8	23.7
	2%	DB	32.8	34.2	34.8	34.1	33.2	32.1	31.0	31.3	31.9	32.1	32.2	32.1
		MCWB	22.1	23.5	23.3	24.3	23.4	23.4	23.2	24.1	24.1	24.2	24.5	23.3
	5%	DB	31.9	33.2	33.8	33.1	32.6	31.1	30.2	30.8	31.0	31.2	31.4	31.3
		MCWB	21.9	23.0	23.4	23.6	23.5	23.0	23.1	23.9	23.9	23.8	24.1	23.0
	10%	DB	31.0	32.2	32.6	32.2	31.6	30.1	29.5	29.8	30.1	30.5	30.6	30.8
		MCWB	22.0	22.7	23.3	23.3	23.4	23.1	23.1	23.5	23.7	23.8	23.9	22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	26.7	26.5	27.2	27.2	26.4	26.0	27.4	26.9	26.9	26.9	26.0
		MCDB	30.3	33.2	32.5	32.9	30.5	30.0	29.6	30.3	30.6	30.4	30.4	30.4
	2%	WB	23.7	25.5	25.4	26.1	26.0	25.2	25.0	25.9	25.7	25.7	25.9	24.9
		MCDB	29.8	32.3	32.0	32.1	30.0	28.9	28.6	29.5	29.2	29.4	30.0	30.3
	5%	WB	23.0	24.7	24.7	25.4	25.3	24.5	24.3	25.0	24.9	25.1	25.3	24.1
		MCDB	29.6	31.0	31.4	30.9	29.4	28.4	28.0	28.4	28.6	29.0	29.4	29.5
	10%	WB	22.4	23.9	24.0	24.6	24.7	23.9	23.8	24.2	24.3	24.5	24.7	23.4
		MCDB	28.9	29.8	30.6	29.6	28.9	27.9	27.4	27.8	28.2	28.5	28.7	29.0
Mean Daily Temperature Range	5% DB	MDBR	12.8	12.6	12.3	10.5	9.2	8.8	9.2	9.3	9.6	9.6	10.2	11.7
		MCDBR	13.4	13.4	13.1	11.5	10.7	10.3	10.3	10.2	10.6	10.3	11.1	12.2
		MCWBR	5.2	5.2	4.9	4.3	4.1	4.2	4.7	4.7	4.5	4.2	4.6	5.2
	5% WB	MCDBR	12.7	12.5	12.1	10.3	9.2	8.8	9.3	9.6	9.8	9.6	10.3	11.3
		MCWBR	5.6	6.0	5.6	4.9	4.8	4.4	4.9	5.4	4.8	4.7	5.1	5.6
Clear-Sky Solar Irradiance	taub	0.363	0.386	0.407	0.469	0.444	0.443	0.439	0.409	0.404	0.390	0.377	0.379	
	taud	2.567	2.482	2.436	2.246	2.358	2.368	2.375	2.475	2.475	2.534	2.572	2.546	
	Ebn at Noon	947	933	913	847	852	843	849	887	903	916	925	922	
	Edh at Noon	100	113	120	143	125	122	122	112	113	105	99	101	
All-Sky Solar Radiation	RadAvg	5.29	5.76	5.79	5.34	5.17	5.29	5.38	5.47	5.55	5.23	4.84	4.90	
	RadStd	0.21	0.28	0.46	0.40	0.31	0.21	0.20	0.22	0.21	0.17	0.33	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.78	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page